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Estimating the effect of hoof health on milk yield in British dairy cattle

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ABSTRACT

Lameness is a highly prevalent and economically important health problem in British dairy cattle. Previous studies have highlighted a delay between lameness onset and farmer detection. This lag, combined with the low sensitivity of clinical diagnosis, limits early intervention, thereby worsening both production and animal welfare outcomes. Notably, the negative effects of lameness can persist even after clinical recovery. Understanding the burden of health issues is essential for informed decision making at the farm level and beyond. This study quantifies the impact of hoof-health disorders on milk yield in British dairy cattle. Data were obtained from 6,050 cows across a nonrandom sample of 11 farms, including repeated daily milk yield records, totaling ~2.1 million cow-days. A mixed linear model was developed to account for the time-dependent, correlated nature of milk yield data, using a first-order autoregressive correlation structure. An event study design allowed investigation of milk yield trajectories before and after clinical diagnosis. Farmer-reported data were used to identify hoof-related lameness events. After adjusting for mastitis, stage of lactation, parity, farm, season, and breed, cows with a hoof-health event produced on average 1.47 kg/d less milk than healthy cows. For severe cases, the average daily reduction reached 2.05 kg. The event study revealed that milk yield declines were detectable before clinical diagnosis and persisted after treatment. These findings underscore the impact of hoof-health disorders on dairy productivity and highlight the need for earlier

detection as it could help farmers mitigate milk losses, reduce treatment costs, and enhance animal welfare.

Key words: hoof disorders, hoof health, dairy cattle, milk yield, impact, United Kingdom

INTRODUCTION

With British dairy cattle exhibiting high levels of hoof-related lameness (Afonso et al., 2020), it is important to understand how this affects animal production and welfare. Lameness is reported to be the second costliest health condition in dairy cattle in the UK (Bennett and Ijpelaar, 2005; Bennett et al., 1999), the impact of which arises from production losses and expenditure in treatment and control (Willshire and Bell, 2009; White, 2011; Bennett et al., 1999; Charfeddine and Pérez-Cabal, 2017; Dolecheck and Bewley, 2018).

Various studies conducted globally have documented the adverse effect of lameness on milk yield (Coulon et al., 1996; Hernandez et al., 2005; Bicalho et al., 2008; Onyiro et al., 2008; Logroño et al., 2021; Puerto et al., 2021). In the United Kingdom, Green et al. (2002) indicated an average loss of 357 kg of milk per lactation due to clinical lameness, and Archer et al. (2010) reported a reduction of 350 kg of milk for a severe case in the first month of lactation over 305 d (Green et al., 2002; Archer et al., 2010). Amory et al. (2008) investigated the milk yield losses according to the underlying lameness cause, reporting a milk loss per lactation in relation to sole ulcer and white line disease at 574 and 369 kg, respectively (Amory et al., 2008).

Hoof-related lameness has been shown to decrease the productivity of the cow before and after disease is clinically perceptible (Green et al., 2010). High-yielding cows appear to be more susceptible to developing lameness (Green et al., 2002; Amory et al., 2008; Bicalho et al., 2008; Archer et al., 2010). Additionally, the

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The list of standard abbreviations for JDS is available at adsa.org/jds-abbreviations-25. Nonstandard abbreviations are available in the Notes.

effect of hoof-related lameness lingers, as recurrence is likely following the first case (Green et al., 2002; Randall et al., 2015).

From a policy and herd management perspective, these observations underscore the need for enhanced strategies to detect and prevent hoof disorders. Investing in early detection technologies, routine mobility scoring, and preventive hoof care can reduce milk losses and animal suffering (Silva et al., 2021; Roche et al., 2024). Demonstrating the tangible production losses attributable to lameness can also help motivate greater investment from farmers in preventive practices, both improving economic performance and promoting animal welfare.

This study aims to contribute to the understanding of the effect of hoof-health disorders in dairy cattle by exploiting a large dataset comprising over 2.1 million daily milk yield observations and of a nonrandom sample of British dairy farms to (1) estimate the effect of hoof-related health events on daily milk yield, (2) conduct an event study approach to assess changes in milk yield before and after the reported hoof-health events, and (3) estimate the effect of specific hoof-related health disorders on daily milk yield.

MATERIALS AND METHODS

The dataset was developed through information shared by a subset of farmers during the questionnaire process of a sample population of dairy farmers for a larger research study that aimed to characterize the farms in terms of their herd, inputs and outputs, hoof-related health management, assurance schemes, prices, and market. This subset of farmers was selected from the study population based on the criteria outlined below.

The sample population was a nonrandom selection of farms across 4 regions in Great Britain—Northeast Wales, South Scotland, Northwest England, and Southwest England. Farms were identified through 3 veterinary hoof service providers: IceRobotics Ltd. (currently part of Peacock Technology Ltd.), Synergy Farm Health Ltd. (SFH), and Shropshire Farm Vets Ltd./Shropshire Hoof Care Ltd. (SFV/SHC). Contact with the farmers was initially established via these health service providers.

Only farms with daily milk yield data at the cow level using UNIFORM-Agri as their farm management software were selected. Consent was obtained to allow researchers access to herd data management software UNIFORM-Agri, including milk recording and health data. A data-sharing protocol was set up with the health service providers and individual farmers. If a farmer changed veterinary service providers during the research process, then contact with the farmer and access to farm data were discontinued. Therefore, data from these farms are not included in this study. More detailed explanation

on the farmers enrollment process, the questionnaire and how data were accessed can be found in the Supplemental Materials (see Notes) and Afonso (2021).

Production and health data concerning hoof health and mastitis at animal level for each individual farm were exported from UNIFORM-Agri to Microsoft Excel (Microsoft Corporation, 2016). Each farm had 3 associated datasets: milk yield, fertility, and health events. Health data were recorded by farmer or animal health caretaker. Datasets were cleaned in Excel to harmonize the health event entries and breed data and then saved in comma-separated values (CSV) format.

There were 26 different breed designations across the cohort of animals. Cows were reclassified into 6 new categories (Ayrshire, British Friesian, Holstein, Holstein-Friesian, Jersey, and Other) according to the uniform breed codes of the National Association of Animal Breeders (National Association of Animal Breeders, 2021), and a new breed variable created. The original breed designation and reclassification can be found in Supplemental Table S3 (see Notes).

There was heterogeneity in how hoof-health events were recorded across the farms. Data were cleaned and reclassified based on the Claw Health Atlas (Egger-Danner et al., 2020) into 5 or 6 different categories according to 2 distinct classification systems. The noninfectious hoof disorders were grouped in 4 categories: white line disease (WLD), sole lesions, toe lesion, and other. The 5-category classification system considered all infectious hoof diseases in one category. The 6-category classification system had bovine digital dermatitis in a category of its own, and the remaining infectious hoof-health diseases were classified under “infectious hoof diseases.” All lesions related to the toe were designated as “toe lesion.” The category “sole lesions” included sole hemorrhage or bruising and sole ulcer among other recorded events. Supplemental Table S4 (see Notes) provides an overview of the health events data and their reclassification.

These CSV files were further cleaned, reshaped, and merged into a single dataset using R (R Core Team, 2020). Several health-related categorical variables were generated: presence of at least one hoof disorder, mastitis-related health event, and specific hoof-related health event (WLD, sole lesion, toe lesion, bovine digital dermatitis (BDD), infectious hoof diseases, other). Each of these was a binary variable equal to 1 if a cow was diagnosed with or treated for the respective hoof-health event on a given day and 0 otherwise. Cows could have multiple hoof disorders on the same day. An indicator for those hoof-health events would be equal to 1. For each day the cows did not have a hoof-health event, the health-related variable was attributed a value of 0.

A binary variable was generated showing whether, on a given day, a cow had been treated by the farmer for

Table 1. Overview of the sample data

Farm	No. of cows	No. of observations	No. of cow-years	Lactation			Daily milk yield		
				Mean (SD)	Minimum	Maximum	Mean (SD)	Minimum	Maximum
1	545	171,704	470.4	2.4 (1.6)	1	11	27.7 (11.0)	0.0	83.6
2	437	200,464	549.2	2.7 (1.6)	1	12	29.4 (11.6)	0.0	96.5
3	706	202,059	553.6	2.5 (1.5)	1	8	32.4 (10.7)	0.0	110.9
4	292	151,496	415.1	2.9 (1.9)	1	10	30.2 (12.0)	0.0	98.7
5	518	187,812	514.6	2.7 (1.4)	1	8	35.1 (14.5)	0.0	80.2
6	241	107,703	295.1	2.8 (1.6)	1	9	36.7 (12.6)	0.0	109.7
7	196	80,941	221.8	3.1 (1.9)	1	9	21.1 (8.2)	0.0	86.3
8	634	224,183	614.2	2.3 (1.5)	1	9	25.5 (14.4)	0.0	85.9
9	380	150,684	412.8	2.6 (1.7)	1	10	29.6 (12.5)	0.0	109.4
10	377	130,047	356.3	2.5 (1.5)	1	9	38.9 (10.6)	0.0	85.3
11	1,724	537,010	1,471.3	2.5 (1.5)	1	8	38.8 (12.2)	0.0	103.2
Total	6,050	2,144,103	5,874.3	2.6 (1.5)	1	12	32.7 (13.3)	0.0	110.9

at least one hoof-related health event, regardless of the type of treatment (e.g., antibiotics, nonsteroidal anti-inflammatory drugs [NSAID] for at least 1 d). The variable took the value of 1 if the animal had been recorded as receiving treatment for at least 1 d and 0 if otherwise. The purpose of this variable is to distinguish mild cases (assuming a value of 0) from more severe cases (assuming a value of 1). This definition was based on the reported treatments, as most cases treated 1 d or more included administration of antibiotics and NSAID. In cases when a cow was diagnosed with a hoof disorder but the number of treatment days was recorded as 0, this indicator took the value of 0. This did not mean that the cow received no treatment for the hoof disorder but rather that no treatment was recorded by the farmer.

An ordinal variable for the lactation cycle was generated. Similar to the approaches of Hanno et al. (2025) and Booth et al. (2004), this categorical variable designates the 60-d period of the lactation cycle in which the milk yield data were recorded (Booth et al., 2004; Hanno et al., 2025). There are 11 categories in total, and the last category of this variable was 600 d in lactation or above. Periods of 60 d were chosen to reflect the biological pattern of the lactation curve and to divide the lactation cycle into periods that follow production trends and disease risk dynamics (Hutjens, 2016). A 4-category ordinal variable for the quarter of the year in which the date of the milk yield data were generated.

The milk yield dataset only included records for the days cows were milked. In other words, if a cow was not milked for any specific reason (e.g., disease, death, drying off), there was no entry in the daily milk data. Although it was known that an animal was not milked, the reason behind it was not shown in the UNIFORM-Agri system. A “dry-off period” was defined as 30+ d with no milk yield record. Rows were added for all other periods of <30 d, and the daily milk yield for these rows

was attributed the value of 0 to bring the data into a panel structure. Attributing a value of 0 (rather than imputing milk yield values) highlights the losses to farmers associated with being unable to milk animals or having to discard the milk of diseased animals. Further information on these gaps in daily milk data are available in the Supplemental Material.

Some cows had missing breed information in the raw data. No attempt was made to find the missing data, and this information was assumed to be unknown and categorized as such. Reproductive data were used to find all calving dates. The first calving date was defined as d 1 of the lactation cycle, after which days increased by 1 every day until the next calving date.

The resulting dataset was an unbalanced panel with milk yield, fertility, and health event data spanning from 10 July 2017 to 30 September 2020 for animals from 11 farms located across 4 regions in Great Britain (North Wales, South Scotland, Northwest England, and Southwest England). These data included: ear tag reference, lactation number (parity), breed, date and time of milking, milk yield recorded in the milkmeter for the corresponding date and time of milking, total daily milk production, DIM, farm, and lastly hoof-health event diagnosis and related parameters (date, duration of illness, number of treatment days, and associated milk withdrawal periods).

Due to its sensitive nature, data are in a private repository and not publicly available. The R code is available upon request. Further details on data format and how the data were cleaned and manipulated can be found in the Supplemental Materials.

Analysis Methods

The magnitude of the reduction in daily milk yield associated with the incidence of a hoof-related health

Table 2. Location and farm management choices of subset of farms used in the analysis

Farm	Location ¹	Calving pattern ²	Heifer replacement strategy	Zero-grazing	Cubicles	Milking Robot	Daily milking frequency	Lameness screening	Foot bathing	Hoof trimming ³
1	Wales NE	AYR	Closed herd	No	Yes	No	2	Yes	Yes	30 d to DO, 60 DPC
2	England SW	AYR	Closed herd	No	Yes	No	2	Yes	Yes	at DO and 80 DPC
3	Scotland S	AYR	Closed herd	No	Yes	No	2	Yes	Yes	at DO, 100 DPC
4	England NW	DB ⁶	Closed herd	No	Yes	No	3	Yes	Yes	30 d to DO, 120 DPC
5	England SW	AYR	Closed herd	No	Yes	Yes	—	Yes	Yes	at DO, 40 DPC
6	England NW	AYR	Open herd	Yes	Yes	Yes	—	Yes	Yes	at DO, 60 DPC
7	England SW	Autumn	Closed herd	No	Yes	No	2	Yes	Yes	at DO
8	Scotland S	AYR	Closed herd	No	Yes	No	2	Yes	Yes	at DO, 80 and 180 DPC
9	England NW	AYR	Open herd	No	Yes	No	2	Yes	Yes	30 d to DO, 60 DPC
10	England SW	AYR	Open herd	No	Yes	Yes	—	Yes	Yes	at DO and 80 DPC
11	England SW	AYR	Closed herd	No	Yes	No	4	Yes	Yes	at DO, 100 DPC

¹NE = Northeast; SW = Southwest; S = South; NW = Northwest.²AYR = all year round; DB = dual block;³DO = dry off; DPC = days post calving.

event was determined by comparing the daily milk yield between groups of cows with and without hoof-health events. The overall impact of any (multiple) hoof-related health events was estimated with the following linear model:

$$Y_{it} = HoofHealthEvent_{it} + Mastitis_{it} + \sum_{j=1}^J LactationStage_{jit} + \sum_{k=1}^K Parity_{ki} + \sum_{l=1}^L Breed_{li} + \sum_{m=1}^M Quarter_{mi} + \sum_{n=1}^N Farm_{ni} + u_i + \varepsilon_{it},$$

Where Y_{it} is the daily milk yield of cow i at time t (expressed in days), u_i is a cow-specific random intercept, and ε_{it} is an AR(1)-structured residual error.

The following variables were included as fixed effects: a binary variable equal to 1 if a cow was diagnosed or treated for any hoof-related health event on day t (and equal to 0 otherwise), a binary variable equal to 1 if a cow was diagnosed or treated for a mastitis-related health event on day t (and equal to 0 otherwise), a full set of dummy variables for the 60-d periods of the lactation cycle (reference category = first 60 d of lactation), a set of dummy variables for parity (including parity 1, 2, 3, and 4+; reference category = parity 1), a full set of dummy variables for breed (reference category = Holstein-Friesian), a full set of dummy variables for farm (reference category = farm 1), a full set of dummy variables for quarter of the year or season (reference category = first quarter). The unique cow ear tag was fitted as individual random effect. All farm-specific, time-invariant characteristics and management practices were captured by the farm fixed effects. It is important to note that this only captures farm management practices (e.g., milking frequency, farming method, farm location, grazing regimen, and so on) that did not change over the sample period.

We estimated a mixed linear model using the REML method. To account for the pattern of correlation between cow-level observations, the approach of Gröhn et al. (1999) was followed and a mixed model with a first-order autoregressive correlation matrix—AR(1)—was used (Gröhn et al., 1999). The statistical and regression analysis was conducted using the R language, libraries, and packages (R Core Team, 2020), particularly with the *lme* function within the *nlme* package (Jose et al., 2013).

We analyzed the effect of hoof-health events on milk loss in a 3-stage approach—first evaluating the average impact resulting from nonspecific hoof disorders (stage 1), then assessing the average impact of nonspecific hoof disorders before and after the moment of diagnosis (stage 2), and finally determining the average effect of specific hoof disorders (stage 3).

As outlined above, stage 1 assessed the average milk loss due to a hoof-health event. We began by utilizing

Table 3. Breed composition of dairy cohort by farm

Farm	No. cows	Ayrshire, n (%)	British Friesian, n (%)	Holstein, n (%)	Holstein-Friesian, n (%)	Jersey, n (%)	Other, n (%)	No breed data, n (%)
1	545	1 (0.2)	22 (4.0)	351 (64.4)	147 (27.0)	12 (2.2)	12 (2.2)	—
2	437	—	—	—	373 (85.4)	—	1 (0.2)	63 (14.4)
3	706	—	—	2 (0.3)	293 (41.5)	1 (0.1)	—	410 (58.1)
4	292	—	—	1 (0.3)	291 (99.7)	—	—	—
5	518	—	6 (1.2)	33 (6.4)	135 (26.1)	—	343 (66.2)	1 (0.2)
6	241	—	—	119 (49.4)	122 (50.6)	—	—	—
7	196	—	1 (0.5)	—	195 (99.5)	—	—	—
8	634	—	15 (2.4)	566 (89.3)	51 (8.0)	—	2 (0.3)	—
9	380	—	—	23 (6.1)	324 (85.3)	—	14 (3.7)	19 (5.0)
10	377	—	—	200 (53.1)	177 (46.9)	—	—	—
11	1,724	—	—	61 (3.5)	1,649 (95.6)	—	6 (0.3)	8 (0.5)
Total	6,050	1 (0.0)	44 (0.7)	1,356 (22.4)	3,757 (62.1)	13 (0.2)	378 (6.2)	501 (8.3)

the cohort of animals that had a recorded hoof-health incident regardless of the severity of illness, and we then used the cohort of more severely diseased animals for which at least 1 d of treatment was recorded (treatment length variable = 1; Warnick et al., 2001; Juarez et al., 2003; Norring et al., 2014; referred to as *Baseline* in the table with the results in the corresponding section).

Acknowledging the importance of distinguishing between a reduction in milk yield due to a disease-related physiological reduction from discarded milk resulting from a hoof-health event, we re-estimated the model with the interaction of a binary variable showing whether diseased cows continued being milked throughout the duration of their illness and the variable indicating the incidence of any hoof-health related event (referred to as *Interaction* in the table with the results in the corresponding section). This variable labeled *Milkmeter* took the value of 1 if there was no milk yield record from milkmeter and 0 if otherwise. This analysis provided insight into the potential reduction in losses associated with early diagnosis and treatment, leading to a complete or partial reduction of discarded milk.

To explore the potential influence of farms with low incidence rates on the results, we conducted a sensitivity analysis by re-estimating the models after excluding the farms in the lowest quartile of incidence rate ($Q1 = 5.65$ cases/100 cows per year). Thus, we repeated the analysis above excluding farms 1, 7, and 9. Additional details on this analysis are available in the Supplemental Materials.

Stage 2 of the analyses looked at the impact of hoof disorders in relation to the moment of diagnosis. The analyses discussed above rely on farmers' diagnostic data to identify the onset of illness; however, researchers have observed a reduction on milk yield before the

time of diagnosis and a slow return to the expected milk production levels even after the end of the treatment period (Lucey et al., 1986; Green et al., 2010). Furthermore, the 305-d milk production estimated from monthly tests usually does not capture short-term fluctuations and drops in milk production (Wilson et al., 2004). Making use of the granularity of our dataset, we applied an event study framework to investigate the impact of hoof-health events before and after the moment of diagnosis.

Prior literature, such as Lucey et al. (1986) and Lischer et al. (2000) among others, show that insults to the animal's hoof may produce effect in reducing milk yield 9 to 12 wk before the clinical signs of disease. Therefore, we began the event study analysis by re-estimating the *baseline* model (with the aforementioned control variables) using a full set of weekly dummy variables for the 12 wk before and after diagnosis instead of the hoof-related health event indicator. That is, a dummy variable was included for every week more than 12 wk before diagnosis and, separately, a dummy variable for every week more than 12 wk after diagnosis, with the week of diagnosis as the reference period.

To gain further insight into the observable reduction in milk yield before and after diagnosis, we then repeated the event study analysis using a full set of dummy variables for the 30 d before and after diagnosis and using the aforementioned control variables. A dummy variable was included for every day more than 30 d before diagnosis and a dummy variable for every day more than 30 d after diagnosis, with the day of diagnosis as reference period. This allowed us to assess whether aggregating data could mask the impact of hoof disorders and to find if there was an observable reduction in milk yield before the day of diagnosis.

Table 4. Distribution of the hoof-related lesions across the sample population according to different parameters

Item	No. of cow-years	No. of hoof-health events	No. of sick days ¹	Average sick days per health event	Incidence per 100 cow-years ²						
					Overall	IHD	WLD	SL	TL	BDD	Other
Total	5,874.3	1,822	2,114	1.16	31.0	2.8	6.7	8.7	0.2	4.9	7.7
Farm											
1	470.4	23	35	1.52	4.9	1.1	0.0	0.2	0.0	1.5	2.1
2	549.2	91	134	1.47	16.6	0.5	5.6	6.6	1.1	0.0	2.7
3	553.6	455	470	1.03	82.2	2.5	35.9	30.9	0.4	9.0	3.4
4	415.1	193	263	1.36	46.5	4.3	22.2	9.2	0.0	4.3	6.5
5	514.6	87	87	1.00	16.9	0.8	0.2	0.8	0.0	0.2	15.0
6	295.1	19	22	1.16	6.4	5.1	0.0	1.0	0.0	0.0	0.3
7	221.8	5	12	2.40	2.3	2.3	0.0	0.0	0.0	0.0	0.0
8	614.2	241	260	1.08	39.2	5.5	5.9	11.7	0.7	6.0	9.4
9	412.8	3	3	1.00	0.7	0.0	0.0	0.0	0.0	0.2	0.5
10	356.3	365	432	1.18	102.4	8.1	9.3	40.1	0.3	29.2	15.4
11	1,471.3	340	392	1.15	23.1	2.5	0.0	3.1	0.0	4.7	12.8
Breed											
Ayrshire	0.8	0	0	—	0.0	0.0	0.0	0.0	0.0	0.0	0.0
British Friesian	39.5	7	7	1.00	17.7	0.0	5.1	2.5	0.0	5.1	5.1
Holstein	1,270.5	421	489	1.15	33.1	4.5	3.4	11.2	0.3	6.5	7.2
Holstein-Friesian	3,723.1	1,002	1,210	1.21	26.9	2.5	4.8	6.8	0.2	4.6	8.0
Jersey	8.1	0	0	—	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Other	381.0	52	52	1.00	13.6	1.0	0.3	0.8	0.0	0.3	11.3
NA	451.5	340	356	1.05	75.3	2.0	36.8	25.2	0.2	6.9	4.2
Parity											
1	1,718.1	372	372	1.18	21.7	2.3	2.7	7.0	0.1	4.8	4.7
2	1,539.4	374	374	1.16	24.3	2.2	4.7	4.7	0.6	5.2	6.8
3	1,076.5	334	334	1.17	31.0	2.5	7.2	8.9	0.0	4.0	8.5
4	727.1	350	350	1.12	48.1	4.0	12.4	14.2	0.1	5.5	12.0
5	483.0	226	226	1.12	46.8	4.6	11.4	13.7	0.4	5.4	11.4
6	205.7	96	96	1.25	46.7	2.9	10.2	18.5	0.0	5.3	9.7
7	74.5	46	46	1.20	61.7	2.7	20.1	18.8	0.0	5.4	14.8
8	35.5	12	12	1.08	33.8	2.8	19.7	5.6	0.0	2.8	2.8
9	10.2	9	9	1.00	88.2	29.4	39.2	9.8	0.0	0.0	9.8
10	3.0	3	3	2.33	100.0	0.0	100.0	0.0	0.0	0.0	0.0
11	1.2	0	0	—	0.0	0.0	0.0	0.0	0.0	0.0	0.0
12	0.3	0	0	—	0.0	0.0	0.0	0.0	0.0	0.0	0.0

¹Includes cases in which animals were diagnosis with a hoof disorder but had 0 d of treatment.

²IHD = infectious hoof diseases (excluding BDD); WLD = white line disease; SL = sole lesions; TL = toe lesions; BDD = bovine digital dermatitis.

In stage 3 of the analyses, we determined the impact of specific types of hoof disorders on daily milk yield. We repeated the first stage analysis using individual variables for specific hoof-related health events instead of an aggregate variable. These hoof-related health events included infectious hoof disease, WLD, sole lesions, toe lesions, and other. It is important to note that BDD has a high incidence rate but a low impact on milk yield (Barkema et al., 1994; Amory et al., 2008; Relun et al., 2013; Afonso et al., 2020). Thus, an analysis using this 5-category classification system may lead to underestimating the true impact of other infectious hoof disorders. Therefore, we repeated this final analysis using a 6-category classification system.

RESULTS

Summary Statistics

The sample subset was composed of 6,050 cows unevenly distributed across 11 farms, ranging from 196 cows (3.2% of the total animals) originating from farm 7 to 1,724 cows (28.5% of the total animals) originating from farm 11 (Table 1). For this cohort of animals, there were around 2.1 million days of milk yield data (or 5,874 cow-years). The milk yield data were recorded from July 10, 2017, to the September 30, 2020. The average lactation number was quite similar across farms, with the lowest mean lactation at 2.3 (1.5 SD)

Table 5. Multivariable regression analysis on the impact of any hoof-health event on daily milk yield (kg) regardless of the duration of treatment¹

Health event	Baseline ²	Interaction ³
Hoof-related	−1.47*** (−1.86, −1.08)	−0.87*** (−1.28, −0.47)
Milkmeter ⁴	—	−1.00*** (−1.50, −0.51)
Hoof-related*No milk	—	−7.01*** (−8.41, −5.61)
Observations	1,750,016	1,750,016

¹95% CI reported in parentheses. All models estimated in R using the lme function from the nlme package with AR(1) correlation structure. Control variables include mastitis, day of lactation cycle, parity, breed, quarter of year, and farm as described in the Materials and Methods section. Complete regression results are available in the Supplemental Material.

²Includes estimates for the impact of any hoof-related event on daily milk yield, irrespective of whether milk was discarded or reduced due to the occurrence of a hoof-health event.

³Includes estimates for the impact of any hoof-related event on daily milk yield including the interaction between discarded milk and occurrence of hoof-health event.

⁴A binary variable = 1 if there was no milk yield record from milkmeter and = 0 otherwise.

* $P < 0.1$; ** $P < 0.05$; *** $P < 0.01$.

at farm 8 and the highest at 3.1 (1.9 SD) at farm 7. The cow with highest lactation number (12) was reported at farm 2. Cows gave on average 32.7 kg (13.3 SD) of milk per day, with a maximum daily milk yield record of 110.9 kg at farm 3. Farms 10 and 11 had the highest mean daily milk yield with 38.9 kg (10.6 SD) and 38.8 kg (12.2 SD), respectively. Farm 7 had the lowest level of production, with cows producing on average 21.1 kg (8.2 SD) of milk per day.

As seen in the summary statistics presented in Table 2, 8 out of 11 farms were in England. Wales was the least represented country with only one farm. Nine out of 11 farms had herd calving all year round. Spring calving systems were not represented in the sample subset. Apart from one, farmers classified the breed of their herds as either Holstein or Holstein-Friesian. Of the 11 farms, 8 had closed herds; the remaining units ran open herds. Only one farm kept its animals indoors all year round, whereas the other 10 allowed the herd to graze during a period of the production cycle. Three farms had milking robots. Of the 8 that had conventional milking systems, 6 milked cows twice a day. All farms kept their herd in cubicles and practiced lameness screening or mobility scoring and foot bathing.

In the total cohort of cows, Holstein and Holstein-Friesian together represented 84.5% of the total number of animals (Table 3). For all farms except farms 3 and 5, most cows across all farms fell in these 2 categories. In farm 5, two-thirds of the cows were classified as “Other” with regards to breed. More than half of the animals (58.1%) from farm 3 did not have data on the breed, whereas the remaining animals were classified as Holstein-Friesian. Even though in a small proportion, Jersey cows could be found in farms 1 and 3, whereas Ayrshire could only be found in farm 1. There were missing data concerning breed for 8.3% of the total

cohort. These animals with missing breed data were primarily located at farm 3.

As seen in Table 4, over the course of the production cycles analyzed, there were 1,822 hoof-related health events requiring a total of 2,114 d of treatment. Each affected cow had on average a treatment length lasting 1.16 d, ranging from 1 d (farms 5 and 9) to 2.4 d (farm 7). The standardized incidence rate of hoof disorders from all causes was 31.0 cases per 100 cows per year. Of the cows for which breed was known, Holstein and Holstein-Friesians had the highest incidence rates of hoof disorders (33.1 and 26.9, respectively). No hoof lesions were recorded for Ayrshire or Jersey cows. The incidence of hoof disorders was particularly high for cows that did not have data on breed (75.3 cases per 100 cows-year). The incidence of hoof lesions increased with lactation number. The most frequent lesions in the cohort of cows were sole lesions, other hoof-related lesions, and WLD (8.7, 7.7, and 6.7 cases per 100 cows-year, respectively). Toe lesions and infectious hoof diseases (excluding BDD) were the least frequent (0.2 and 2.8 cases per 100 cows-year, respectively). Overall incidence of BDD was 4.9 cases per 100 cows-year.

Impact of a Hoof-Health Event on Milk Yield

The results in Table 5 indicate the average impact of any hoof-related event on milk yield considering the total cohort of animals (a complete table of results can be found in Supplemental Table S6). Column *Baseline* includes the estimates for the impact of any hoof-related event on daily milk yield, and Column *Interaction* additionally includes the estimates for the interaction between the occurrence of a hoof-related health event and discarded milk. Relative to a healthy cow (without a hoof problem), the occurrence of

Table 6. Multivariable regression analysis on the impact of any hoof-health event on daily milk yield (kg) for cows that had a treatment record of 1 d or longer (treatment length = 1)¹

Health event	Baseline ²	Interaction ³
Hoof-related	-2.05*** (-2.52, -1.57)	-1.18*** (-1.69, -0.67)
Milkmeter ⁴	—	-1.00*** (-1.50, -0.50)
Hoof-related*No milk	—	-7.03*** (-8.49, -5.57)
Observations	1,750,016	1,750,016

¹95% CI reported in parentheses. All models estimated in R using the lme function from the nlme package with AR(1) correlation structure. Control variables include mastitis, day of lactation cycle, parity, breed, quarter of year, and farm as described in the Materials and Methods section. Complete regression results are available in the Supplemental Material.

²Includes estimates for the impact of any hoof-related event on daily milk yield, irrespective of whether milk was discarded or reduced due to the occurrence of a hoof-health event.

³Includes estimates for the impact of any hoof-related event on daily milk yield including the interaction between discarded milk and occurrence of hoof-health event.

⁴A binary variable = 1 if there was no milk yield record from milkmeter and = 0 otherwise.

* $P < 0.1$; ** $P < 0.05$; *** $P < 0.01$.

any hoof-related health event led to an average loss of 1.47 kg (95% CI: 1.08, 1.86, $P < 0.001$) of milk per day, when controlling for mastitis, period of the lactation cycle, farm, parity, season, and breed. These losses were the result of both discarded milk and a physiological reduction in milk production in diseased cows. When considering the interaction with *milkmeter* (Column *Interaction*) we found that, on average, diseased cows produced 0.87 kg (95% CI: 0.47, 1.28, $P < 0.001$) per day less relative to the nondiseased cohort. In comparison, there was an average loss in daily milk production of 7.88 kg (95% CI: 6.54, 9.22, $P < 0.001$) associated with discarded milk.

When we repeated the analysis excluding the lowest quartile of farms in terms of incidence rate of hoof-related illnesses, the results were consistent with respect to both the magnitude and statistical significance of the coefficients. These results are reported in the supplementary materials in Supplemental Table S7 (see Notes).

The results on the analysis on the cohort of cows that required at least 1 d of treatment are reported in Table 6 (a complete table of results can be found in Supplemental Table S8, see Notes). We found that a cow afflicted with a more severe hoof-related health event produced, on average, 2.05 kg (95% CI: 1.57, 2.52, $P < 0.001$) less milk per day compared with a nondiseased cow. When we repeated the analysis interacting the occurrence of a hoof-related illness with discarded milk, we found that, on average, the more severely diseased cows produced 1.18 kg (95% CI: 0.67, 1.69, $P < 0.001$) per day less than the nondiseased cohort and that there were losses of 8.21 kg per day (95% CI: 6.84, 9.58, $P < 0.001$) associated with discarding the milk of this more severely diseased cohort (Table 6).

The Dynamics of Hoof-Health Events on Milk Yield

The results on the impact on milk yield before and after the moment of diagnosis are illustrated in Figures 1 and 2. The reference categories are the week of diagnosis and the day of diagnosis, respectively.

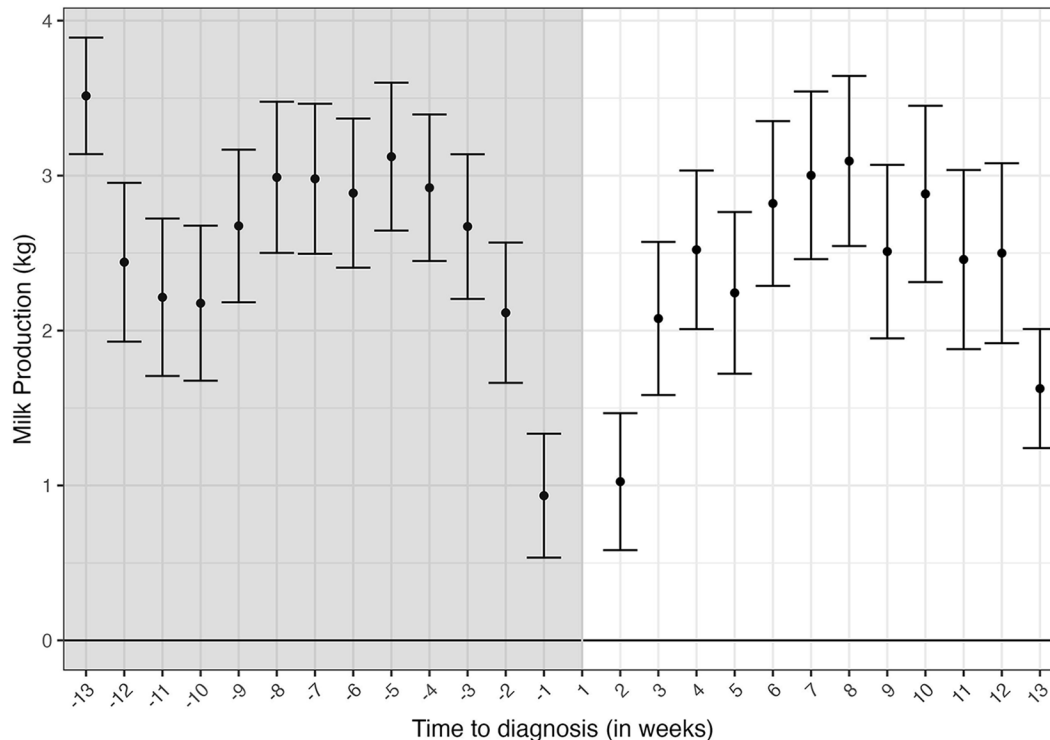
Weekly Milk Yield

The week of diagnosis was associated with the lowest milk yield, as illustrated in Figure 1. However, reduction in milk yield could be seen in the week before diagnosis compared with the previous weeks before diagnosis. More specifically, cows produced around 2.5 to 3 kg of milk per day more from 2 to 13 wk before diagnosis compared with the week of diagnosis, but they only produced around 1 kg/d more during the week before diagnosis. Similarly, milk yield was 1 kg/d more 1 wk after diagnosis. Milk yield returned to pre-illness levels 2 wk after diagnosis (wk 3 in Figure 1).

Daily Milk Yield

A similar pattern in milk yield arose when analyzing milk yield on a daily basis (Figure 2); in addition, we could more precisely identify when a decline in milk yield was observed. Specifically, average daily milk production was around 3 to 4 kg/d higher from 11 to 30 d before diagnosis compared with the day of diagnosis. A small reduction in milk yield could be seen some 10 d before the day of the diagnosis; however, the reduction was more marked in the 5 d before disease presence was observed.

In the fifth, fourth, and third days before diagnosis, cows were producing ~1.5 to 2 kg of milk per day more



Notes: Point estimates show the mean daily milk yield differences for each week before and after diagnosis, with the week of diagnosis as the reference period. Whiskers represent the 95% confidence intervals.

Figure 1. Estimated weekly milk yield relative to the week of diagnosis. Point estimates show the mean daily milk yield differences for each week before and after diagnosis, with the week of diagnosis as the reference period. Whiskers represent the 95% CI.

than on the day of diagnosis. In the 2 d before diagnosis, cows were producing less than 1 kg/d more. Daily milk yield did not increase again until 3 d after diagnosis. The animal returned to the pre-illness levels of milk production 10 d after the day of the diagnosis (Figure 2).

Impact of Specific Types of Hoof-Health Events

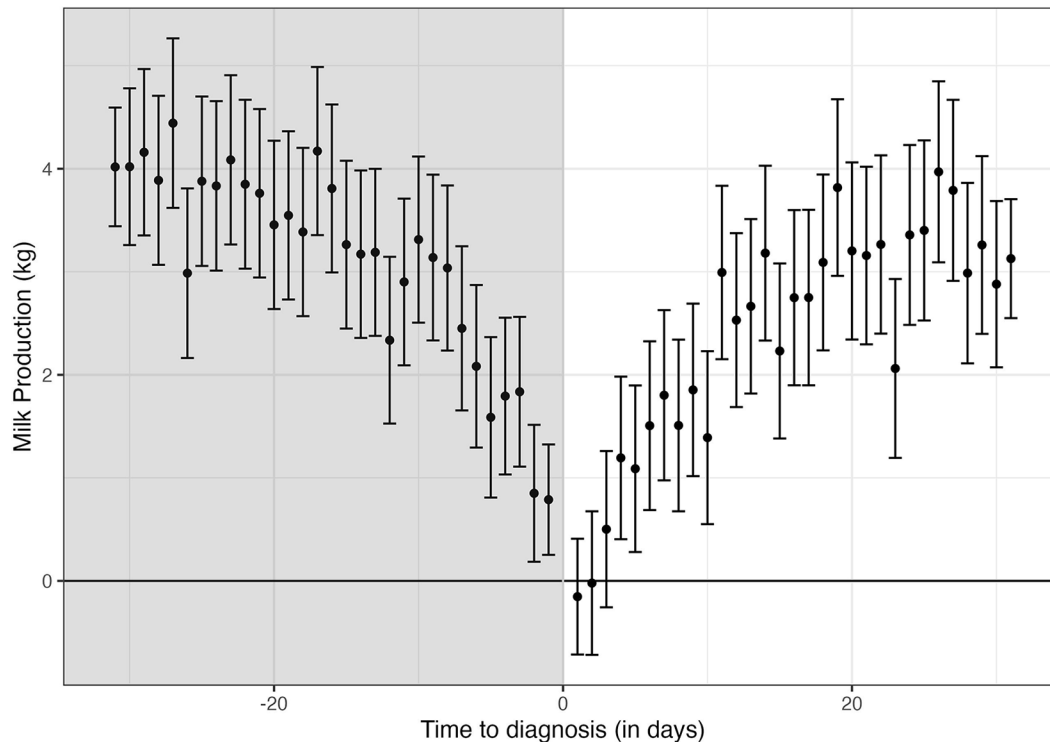
The results on the impact on milk yield from cause-specific hoof-health event are presented in Table 7. We found that only 2 specific hoof-health events are associated with a decrease in milk production: animals afflicted by infectious hoof disease (BDD included) and other hoof problems had an average daily milk loss of 1.66 kg (95% CI: 0.95, 2.37, $P < 0.001$) and 1.91 kg (95% CI: 1.22, 2.61, $P < 0.001$) respectively, when compared with cows without a hoof disorder. When distinguishing BDD from other infectious hoof disease it was observed that, compared with a cow without a hoof disorder, BDD was associated with a daily milk loss of 0.79 kg (95% CI: 0.12, 1.70, $P = 0.089$), whereas other infection hoof diseases led to an average daily loss of 2.63 kg (95% CI: 1.48, 3.78, $P < 0.001$). Other hoof problems were

associated with a loss of 1.93 kg (95% CI: 1.23, 2.63, $P < 0.001$) per day (Table 7).

We repeated the analysis with the cohort of animals assumed to be more severely affected by hoof disorders (received treatment for 1 or more days). As before, we found an increase in the magnitude of the impact of hoof disorders with a similar pattern in terms of statistical significance across the different hoof disorders (Table 8).

DISCUSSION

We estimated an average daily milk loss of 1.47 kg associated with a hoof-health event (95% CI: 1.08, 1.86), which is consistent with several published studies (Warnick et al., 2001; Reader et al., 2011; Navarro et al., 2013). These findings are not sensitive to the exclusion of the farms with low incidence rates. However, there is some (albeit small) evidence that these farms may be underreporting minor illnesses. This highlights the importance of accurate, detailed, and consistent reporting of data across farms. Improved data reporting would allow future research to provide an unbiased estimate of the true losses associated with hoof disorders.



Notes: Point estimates show the mean daily milk yield differences for each day before and after diagnosis, with the day of diagnosis as the reference period. Whiskers represent the 95% confidence intervals.

Figure 2. Estimated daily milk yield relative the day of diagnosis. Point estimates show the mean daily milk yield differences for each day before and after diagnosis, with the day of diagnosis as the reference point. Whiskers represent the 95% CI.

We also separately estimated the impact of more severe hoof disorders, as we assumed these required longer treatments (1 d or more). For these more severely diseased animals, we saw a more pronounced daily milk yield loss of 2.05 kg (95% CI: 1.57, 2.52). When using the interaction term with *Milkmeter* to distinguish the losses associated with discarded milk, we found a reduction in milk production of 0.87 kg/d (1.18 kg/d for the more severely diseased animals and losses associated with the discarding of milk of 7.88 kg/d (95% CI: 6.54, 9.22; 8.21 kg/d for the more severely diseased animals).

Furthermore, the nature of the data allowed us to investigate the impact of hoof-related health events on milk yield with finer granularity, as most studies tend to rely on average weekly or monthly milk yield data (Lucey et al., 1986; Green et al., 2010). Importantly, our event study found that the negative effect of hoof disorders on milk yield can be seen before and after the ailment is diagnosed by the farmer or animal healthcare provider.

The event study findings highlight the importance of early detection of hoof-health problems, as this can help farmers alleviate the impact of these health conditions on

their farms. Early detection and prompt treatment may help reduce production losses as the duration of illness (from onset and not diagnosis of illness) and potentially severity can be reduced. Also, depending on the underlying hoof-related lesions, treatment costs might be eased, further reducing the impact of these health problems. For example, a sole bruise can often be treated solely with hoof trimming, whereas a more severe lesion (e.g., sole ulcer) might imply treatment with antibiotics in case of secondary bacterial infection. Thus, a farmer that identifies a sole bruise before it develops into a sole ulcer avoids a more severe illness with a longer recovery period and higher treatment cost.

The event study also unveils the usefulness of monitoring production traits such as milk yield as screening tool for flagging animals that are potentially underperforming, triggering clinical investigation. The myriad of information captured by farm management systems combined with data science could improve decision making at farm level. The value of such approach in animal health management has been highlighted and presents itself as tool for addressing the complex issues around

Table 7. Multivariable regression analysis of the impact of specific hoof-health events on daily milk yield (kg) regardless of the duration of treatment¹

Health event	5-category classification (BDD within IHD)		6-category classification (BDD outside IHD)	
	Baseline ²	Interaction ³	Baseline ²	Interaction ³
IHD	-1.66*** (-2.37, -0.95)	-1.17*** (-1.92, -0.42)	-2.63*** (-3.78, -1.48)	-1.82*** (-3.09, -0.55)
Other	-1.91*** (-2.61, -1.22)	-1.06*** (-1.80, -0.32)	-1.93*** (-2.63, -1.23)	-1.11*** (-1.85, -0.37)
Sole lesions	-0.28 (-1.02, 0.46)	0.02 (-0.74, 0.77)	-0.30 (-1.04, 0.44)	0.01 (-0.75, 0.76)
Toe lesions	0.52 (-3.29, 4.32)	2.51 (-1.71, 6.73)	0.52 (-3.29, 4.33)	2.52 (-1.70, 6.74)
WLD	-0.12 (-1.08, 0.85)	0.13 (-0.89, 1.14)	-0.07 (-1.03, 0.90)	0.18 (-0.84, 1.19)
BBD	—	—	-0.79* (-1.70, 0.12)	-0.62 (-1.54, 0.30)
Milkmeter ⁴	—	-1.00*** (-1.50, -0.51)	—	-1.03*** (-1.53, -0.54)
IHD* Milkmeter	—	-5.82*** (-8.33, -3.32)	—	-4.40*** (-7.34, -1.45)
Other* Milkmeter	—	-8.45*** (-10.72, -6.18)	—	-8.26*** (-10.56, -5.96)
Sole lesions* Milkmeter	—	-10.60*** (-14.59, -6.61)	—	-10.60*** (-14.59, -6.61)
Toe lesions* Milkmeter	—	-11.68** (-21.45, -1.91)	—	-11.69** (-21.46, -1.92)
WLD* Milkmeter	—	-3.36* (-6.79, 0.07)	—	-3.41* (-6.84, 0.02)
BBD* Milkmeter	—	—	—	-11.96*** (-18.69, -5.23)
Observations	1,750,016	1,750,016	1,749,806	1,749,806

¹95% CI reported in parentheses. All models estimated in R using the lme function from the nlme package with AR(1) correlation structure. IHD = infectious hoof disease; WLD = White line disease; BBD = Bovine digital dermatitis. Control variables include mastitis, day of lactation cycle, parity, breed, quarter of year, and farm as described in the Materials and Methods section. Complete regression results are available in the supplementary material.

²Includes estimates for the impact of any hoof-related event on daily milk yield, irrespective of whether milk was discarded or reduced due to the occurrence of a hoof-health event.

³Includes estimates for the impact of any hoof-related event on daily milk yield including the interaction between discarded milk and occurrence of hoof-health event.

⁴A binary variable = 1 if there was no milk yield record from milkmeter and = 0 otherwise.

* $P < 0.1$; ** $P < 0.05$; *** $P < 0.01$.

dairy herd management in the present and future to come (Egger-Danner et al., 2019; Ezanno et al., 2021).

Study Limitations

A key concern is that of selection bias associated with the enrollment of farmers. Furthermore, these data were based on farm records, a data source that systematically underestimates the problem—literature has indicated that farmers tend to underestimate the magnitude of the lameness problem in their herd or do not maintain accurate records of these health problems due to resource constraints (Whay et al., 2003; Leach et al., 2010; Fabian et al., 2014; Barkema et al., 2015). Thus, the true impact of lameness in British dairy cattle is likely to be higher than the one estimated in this sample population. The eventual underestimation of the impact of hoof disorders is also exacerbated by the health data recording system.

The heterogeneity of disease classification across the different farms and the underreporting problem concerning hoof disorders by dairy farmers, as highlighted in scientific literature, might have hampered the ability to accurately assess the impact of hoof health events on milk production, especially as the impact of hoof-related lameness is cause specific. We observe some evidence of this issue in the sample data. The overall yearly incidence of hoof problems in the cohort was 31.0 cases

per 100 cows, with some farms reporting exceptionally low levels of disease (as low as 0.7 cases/100 cows per year). This incidence rate is somewhat lower than the overall incidence of 36.8 cases of lameness/100 cows per year estimated by Afonso, et al. (2020). Looking at the yearly incidence according to lesion type, the difference is more marked, with farmers having reported at least 40 cases/100 cows per year for each specific hoof lesion in the same study (Afonso et al., 2020). This could reflect selection bias associated with the enrollment of farmers: Those agreeing to participate in the study could have higher quality farm management practices. It could also be due to systematic differences in the recording of these specific hoof events between farms—a problem not unique to this study.

Another aspect to note is that we have only controlled for mastitis as a health-related confounder when estimating the impact of hoof disorders in milk yield. Given that other diseases can lead to milk yield reduction and are also associated with hoof-health (Van Dorp et al., 1998; Rajala-Schultz et al., 1999; Raboisson et al., 2014), if they happen to occur concomitantly, our findings could be a biased estimate of the true effect of hoof disorders. At the time of writing, data on health issues other than mastitis and hoof-related disorders were not available.

As stated in the methods, milk yield dataset had records for the days on which cows were milked. To bring

Table 8. Multivariable regression analysis of the impact of specific hoof-health events on daily milk yield (kg) for cows that had a treatment record of 1 d or longer (treatment length = 1)¹

Health event	5-category classification (BDD within IHD)		6-category classification (BDD outside IHD)	
	Baseline ²	Interaction ³	Baseline ²	Interaction ³
IHD	-2.31*** (-3.17, -1.46)	-1.62*** (-2.53, -0.71)	-3.45*** (-4.66, -2.23)	-2.69*** (-4.05, -1.33)
Other	-3.30*** (-4.28, -2.33)	-1.68*** (-2.77, -0.59)	-3.34*** (-4.31, -2.36)	-1.78*** (-2.88, -0.69)
Sole lesions	-0.82* (-1.75, 0.10)	-0.27 (-1.23, 0.68)	-0.85* (-1.78, 0.07)	-0.29 (-1.24, 0.66)
Toe lesions	-3.03 (-7.66, 1.60)	-0.68 (-6.12, 4.76)	-3.03 (-7.66, 1.60)	-0.68 (-6.12, 4.76)
WLD	-1.41** (-2.51, -0.31)	-1.13* (-2.29, 0.04)	-1.37** (-2.46, -0.27)	-1.08* (-2.24, 0.08)
BDD	—	—	-0.98* (-2.08, 0.12)	-0.66 (-1.78, 0.45)
Milkmeter ⁴	—	-1.00*** (-1.50, -0.51)	—	-1.03*** (-1.53, -0.54)
IHD* Milkmeter	—	-6.03*** (-8.70, -3.36)	—	-3.60** (-6.62, -0.58)
Other* Milkmeter	—	-7.82*** (-10.24, -5.41)	—	-7.58*** (-10.02, -5.15)
Sole lesions* Milkmeter	—	-10.31*** (-14.34, -6.28)	—	-10.29*** (-14.33, -6.26)
Toe lesions* Milkmeter	—	-8.50 (-18.85, 1.86)	—	-8.50 (-18.86, 1.86)
WLD* Milkmeter	—	-2.45 (-6.02, 1.12)	—	-2.50 (-6.07, 1.07)
BDD* Milkmeter	—	—	—	-16.95*** (-24.72, -9.17)
Observations	1,750,016	1,750,016	1,749,806	1,749,806

¹95% confidence intervals reported in parentheses. All models estimated in R using the lme function from the nlme package with AR(1) correlation structure. IHD = Infectious hoof disease; WLD = White line disease; BDD = Bovine digital dermatitis. The control variables include mastitis, day of lactation cycle, parity, breed, quarter of year, and farm as described in the Materials and Methods section. Complete regression results are available in the Supplemental Material.

²Includes estimates for the impact of any hoof-related event on daily milk yield, irrespective of whether milk was discarded or reduced due to the occurrence of a hoof-health event.

³Includes estimates for the impact of any hoof-related event on daily milk yield including the interaction between discarded milk and occurrence of hoof-health event.

⁴A binary variable = 1 if there was no milk yield record from milkmeter and = 0 otherwise.

* $P < 0.1$; ** $P < 0.05$; *** $P < 0.01$.

the data into a panel structure, the absence of milk yield data was considered as either a dry-off period (30+ d with no milk yield record) or as milk withdrawal. However, there could be alternative reasons for the lack of milk data such as misread eartag, milkmeter malfunction, or illnesses other than hoof disorders and mastitis. This kind of error would result in an underestimate of the true impact of hoof-related disorders on daily milk yield. We explore this potential issue in detail in the Supplemental Material.

The representativeness of our data is also a concern; however, the sample is consistent with the official data reported by AHDB and DEFRA with regards to some parameters. For example, around 3 quarters of the sample was located in England and Wales, approximately the same proportion as the one reported by AHDB in 2020 (Afonso, 2021). Approximately 70% of the sample had an aligned milk supply contract with either Arla or Muller, which seems to be in line with the market share of these 2 companies according to anecdotal data. On the contrary, all-year-round calving systems were represented to a lesser extent and autumn calving were more prevalent compared with official data (AHDB, 2017). The sampling strategy failed to recruit smaller herd farms: roughly 75% of British dairy producers milk fewer than 150 cows

(AHDB, 2019), but only 25% of the herds in the sample had fewer than 150 animals. However, it must be said that although small herds are highly prevalent, they account for only a small proportion of the total national herd, as 82.6% of the national dairy cattle population is farmed in units with herds larger than 100 animals (AHDB, 2019). Furthermore, farm survey publications estimate that 78% to 90% of British dairy cattle are Holstein and Friesian breeds (King et al., 2020), whereas in the sample only 72.4% of the cows were of such breeds. Regardless of the characteristics of our sample in relation to survey data for the dairy sector in the UK, results from the study should be valid for the sample population considered, and any extrapolation to the wider population should be conducted with caution, particularly given the nonrandom nature of the sample.

It is important to note that organic farms were intentionally excluded from the study. Given the small proportion of livestock reared in organic farms (3.2% in 2020 according to a report by DEFRA; Office for National Statistics, 2021) and the fact these systems have distinct farming practices when compared with nonorganic farming, a decision was made to concentrate data collection efforts in systems that were consistent to the most common British dairy farm.

CONCLUSIONS

In the cohort, a hoof-health event led to a daily milk loss of 1.47 kg. The impact of the lesions assumed to be more severe was set at 2.05 kg of milk/d. Additionally, the negative effect of a hoof disorder on milk production could be seen 2 wk before and after the week of diagnosis, shedding light on the unperceived milk losses due to hoof disorders. These findings reinforce the need for better prevention strategies through both early identification and treatment.

NOTES

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Nonstandard abbreviations used: BDD = bovine digital dermatitis; CSV = comma-separated values; DE-FRA = XXX; IHD = infectious hoof disease; NSAID = nonsteroidal anti-inflammatory drugs; SL = sole lesions; TL = toe lesions; WLD = white line disease.

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